LAMPS/LIGHTING - EXTERIOR (Continued)

OPERATION - TURN SIGNAL & HAZARD WARNING SYSTEM

The turn signal system operates on battery voltage received on a ignition switch output (run) circuit so that the turn signals will only operate with the ignition switch in the On position. The hazard warning system operates on non-switched battery voltage so that the hazard warning remains operational regardless of the ignition switch position. When the turn signal switch control stalk is moved up or down, the turn signal system is activated. When the turn signal system is activated, the circuitry of the turn signal switch and the combination flasher will cause the selected turn signal indicator, front park/turn signal lamp, front side marker lamp, repeater lamps, and rear tail/stop/turn signal lamp to flash on and off. When the hazard warning system is activated, the circuitry of the hazard warning switch and the combination flasher will cause both the right side and the left side turn signal indicators, front park/turn signal lamps, front side marker lamps, repeater lamps and rear tail/stop/turn signal lamps to flash on and off.

In order to provide the turn signal-on warning, the ElectroMechanical Instrument Cluster (EMIC) monitors vehicle speed and distance messages received from the PCM over the PCI data bus and the hard wired turn signal switch input to the cluster electronic circuit board. If a turn signal remains indicated for a distance of greater than about 1.6 kilometers (1 mile) and the vehicle speed remains greater than about 24 kilometers-per-hour (15 miles-per-hour), the EMIC generates a repetitive chime at a slow rate to provide an audible reminder that a turn signal has been left on (distance is slightly longer with export vehicles). Once the warning chime begins to sound, it will continue until the turn signal is cancelled, until the vehicle speed falls below about 24 kilometers-per-hour (15 miles-per-hour), or until the ignition switch is turned to the Off position, whichever occurs first. This feature is not activated by a hazard warning input to the instrument cluster.

During both the turn signal and the hazard warning operation, if the exterior lamps are turned Off, the front park/turn signal lamps, repeater lamps and the front side marker lamps will flash in unison. If the exterior lamps are turned On, the front park/turn signal lamps, repeater lamps and the front side marker lamps will flash alternately.

WARNING

WARNING:: EYE PROTECTION SHOULD BE USED WHEN SERVICING GLASS COMPONENTS. PERSONAL INJURY CAN RESULT.

CAUTION: Do not touch the glass of halogen bulbs with fingers or other possibly oily surface, reduced bulb life will result.

CAUTION: Do not use bulbs with higher candle power than indicated in the Bulb Application table at the end of this group. Damage to lamp and/or Daytime Running Lamp Module can result.

CAUTION: Do not use fuses, circuit breakers, or relays having greater amperage values than indicated on the fuse panel or in the Owner's Manual.

NOTE: When it is necessary to remove components to service another, it should not be necessary to apply excessive force or bend a component to remove it. Before damaging a trim component, verify hidden fasteners or captured edges are not holding the component in place.

DIAGNOSIS AND TESTING

DIAGNOSIS AND TESTING - TURN SIGNAL & HAZARD WARNING SYSTEM

Always begin any diagnosis by testing all of the fuses and circuit breakers in the system. Refer to the appropriate wiring information.

When diagnosing the turn signal or hazard warning circuits, remember that high generator output can burn out bulbs rapidly and repeatedly. If this is a problem on the vehicle being diagnosed (Refer to 8 - ELECTRICAL/CHARGING - DIAGNOSIS AND TESTING). If the problem being diagnosed is related to a failure of the turn signals to automatically cancel following completion of a turn, inspect the multifunction switch for a faulty or damaged cancel actuator and inspect the turn signal cancel cam lobes on the clockspring mechanism for damage or improper installation. For complete circuit diagrams, refer to the appropriate wiring information.